Gwendal Beaumont

PhD student in Computer Sciences

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Education

2023 – **PhD — Computer Sciences**, *IMT Atlantique*, Brest, France
Title of the thesis: TwinDevOps, an automated process for the deployment and dynamic evolution of Digital Twins.

2020 – 2023 **Engineer's Degree — General Engineering**, *IMT Mines Alès*, Alès, France
One and a half year of general training and one and a half year in the Department of Software Engineering and Artificial Intelligence.

2018 – 2020 **Classe préparatoire aux grandes écoles**, *Lycée Malherbe*, Caen, France
Two-years intensive training to prepare for the French engineering schools competitive exam. Fields of study: Mathematics, Physics, Computer Sciences.

Experience

2026-01 – 2026-04 **Doctoral Visiting Researcher**, *Technische Universität Wien*, Vienna, Austria
Doctoral mobility hosted by Assoc. Prof. Bork Dominik where I worked on an extansion of a process developped during my PhD thesis.

2023-03 – 2023-09 **Intern**, *ALTEN*, Rennes, France
Research & Development internship on the labelling process of data in order to be used with Machine Learning algorithms. Detailed achievements:
- Creation of a framework using Java Spring Boot to manage unlabelled data and get statistics from it
- Creation of an interface using TypeScript and the Electron framework

2022-05 – 2022-08 **Visiting International Scholar**, *Miami University*, Oxford, OH, United States of America
Internship as an assistant in the preparation of a master thesis on the effect of errors in agent-based simulation systems. Detailed achievements:
- Creation of a simulation framework in Python to take errors into account
- Creation of multiple agent-based simulations in Python to test our framework

Services

2026 **Seminar co-organisation**, *IMT Atlantique*, Brest, France
I co-organised one day of the 2026 Annual Seminar of the Department of Computer Sciences. Planning of the seminar is available here.

2024 – 2025 **PhD students representative at the Student Board**, *EULiST Alliance*
Institut Mines-Télécom PhD students representative at the european level in the EULiST Alliance. This alliance integrates over 200,000 students and 20,000 staff at 10 universities in 10 countries.

2023 – 2025 **PhD students representative in a workgroup**, *Institut Mines-Télécom*, France
IMT PhD students representative at the national level in a work group focused on respect and equality.

2023 – 2025 **PhD students representative at the board of directors**, *Institut Mines-Télécom*, France
IMT PhD students representative at the national level for two years. I took part at the board of directors and worked with the scientific director of Institut Mines-Télécom on ways to improve PhD environment.

2021 – 2022 **Communication Manager**, *IMT Mines Alès student council*, Alès, France
Member of the communication managers team for the student council during a year. I administrated the student council's website, managed social networks, and answered and moderated emails and comments on social networks.

Languages

French	Native	English	Fluent *(TOEIC: 975/990)*
Spanish	Intermediate	German	Basics
Korean	Basics		

Conferences

[1] Gwendal Beaumont et al. "Engineering Digital Process Twins: Towards a Methodology Based on MDE and DevOps". In: *Proceedings of the ACM/IEEE 29th International Conference on Model Driven Engineering Languages and Systems*. MODELS Companion '26. Málaga, Spain: Association for Computing Machinery, Oct. 2026.

[2] Gwendal Beaumont et al. "Vers l'automatisation de la gestion du cycle de vie des jumeaux numériques". In: *Actes de la 44ème édition de la conférence INFORSID*. INFORSID 2026. Toulouse, France, May 2026, pp. 55–57. URL: `https://inforsid2026.sciencesconf.org/data/Actes_INFORSID2026.pdf`.

[3] Gwendal Beaumont et al. "Towards Automating the Life Cycle Management of Digital Twins". In: *Conceptual Modeling*. Ed. by Dominik Bork et al. Vol. 16189. Springer Nature Switzerland, Oct. 2025, pp. 412–430. ISBN: 978-3-032-08623-5. DOI: `10.1007/978-3-032-08623-5_22`.

[4] Gwendal Beaumont et al. "Towards Re-Engineering Digital Twins: Preliminary Experiments on Three Use Cases". In: *Proceedings of the ACM/IEEE 27th International Conference on Model Driven Engineering Languages and Systems*. MODELS Companion '24. Linz, Austria: ACM, Sept. 2024, pp. 453–458. ISBN: 979-8-4007-0622-6. DOI: `10.1145/3652620.3688259`.

[5] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. "On the Necessity of Human Decision-Making Errors to Explain Vaccination Rates for Covid-19: An Agent-Based Modeling Study". In: IEEE Computer Society, May 2023, pp. 413–424. URL: `https://ieeexplore.ieee.org/document/10155377`.

Journals and Book Chapters

[1] Gwendal Beaumont et al. "DevOps for and by Digital Twins Leveraging Virtual Replicas in Continuous Software Engineering". In: (June 2026). Ed. by Bedir Tekinerdogan and Cor Verdouw, pp. 227–242. DOI: `10.1016/B978-0-443-45573-5.00018-5`.

[2] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. "A Framework for the Comparison of Errors in Agent-Based Models Using Machine Learning". In: *Journal of Computational Science* 72 (Sept. 2023), p. 102119. ISSN: 1877-7503. DOI: `10.1016/j.jocs.2023.102119`.

[3] Jack T. Beerman, Gwendal Beaumont, and Philippe J. Giabbanelli. "A Scoping Review of Three Dimensions for Long-Term COVID-19 Vaccination Models: Hybrid Immunity, Individual Drivers of Vaccinal Choice, and Human Errors". In: *Vaccines* 10.10 (Oct. 2022), p. 1716. ISSN: 2076-393X. DOI: `10.3390/vaccines10101716`.